An Overview of Deep Space Navigation



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Agenda

- **Background**
- **Mission Design**
  - Planning the route
- **Navigation**
  - Executing the plan
- **Examples**
  - Mars2020
  - Deep Impact

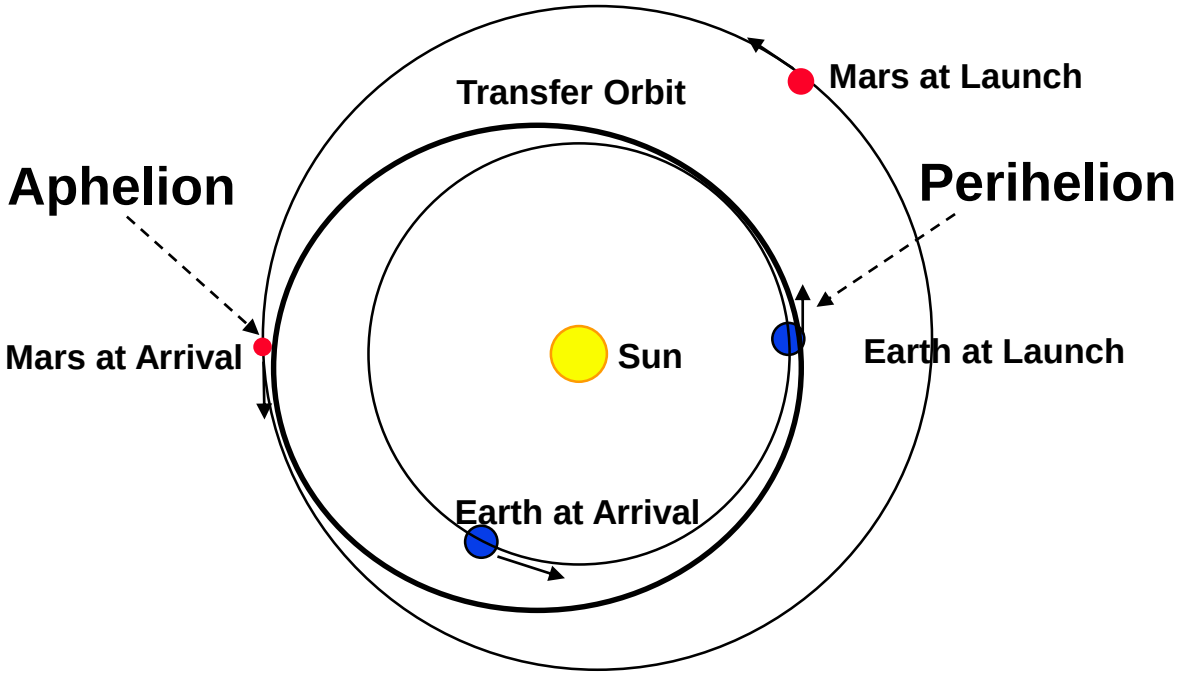
Background

- **Why "deep space" navigation?**
  - To distinguish from navigation in Earth orbit
- **Characteristics of Earth orbit navigation**
  - Primarily done by GPS
  - Geometrically rich data, almost always available
    - GPS receiver provides real time position solutions at rates greater than 1 Hz
    - Can use this or raw observations to compute position and velocity of spacecraft
  - Coverage global in low Earth orbit, can be used out to Geosynchronous orbit and slightly beyond
- **Characteristics of deep space navigation**
  - Primarily done with antennas of the Deep Space Network (DSN) located in Goldstone (CA), Madrid (Spain), and Canberra (Australia)
    - Occasionally augmented by stations in Usuda Japan (ISAS), and Perth Australia (ESA)
  - Used for missions from cis-lunar space to beyond the solar system, including spacecraft which orbit other bodies in the solar system
  - Geometrically poor, sometimes sparse data which provide only single dimensional measurements to infer multi-dimensional state

Mission Design

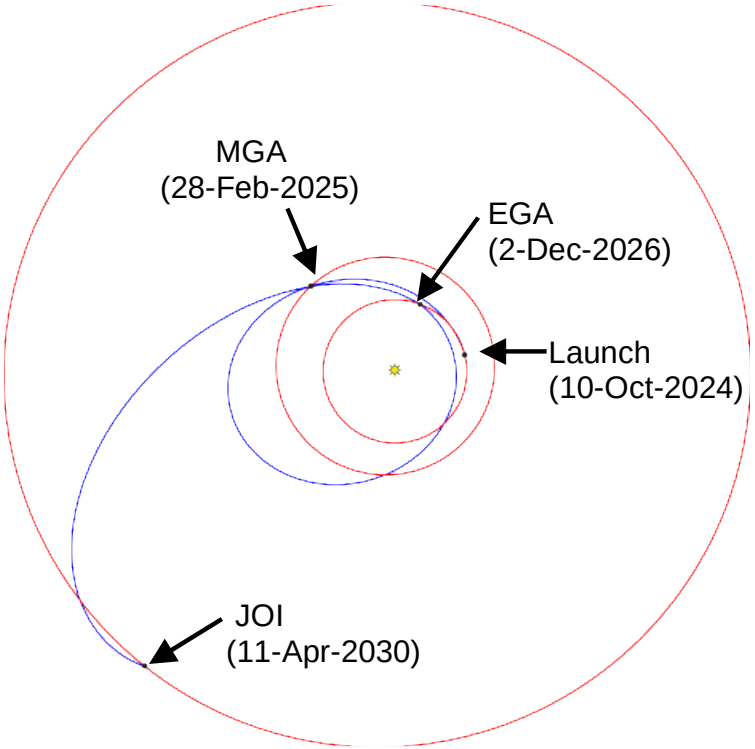
- **Design mission to achieve scientific objectives taking into account operational constraints on the spacecraft**
  - Analogous to planning the route on a road map, or charting a course across the ocean for a ship
- **The "map" in this case is a mathematical model of a spacecraft's trajectory in the solar system, taking into account all the forces which act on the spacecraft**
  - Gravitational forces on the spacecraft
    - Central body force (dominant body is treated as spherically symmetric; this produces pure Keplerian motion)
    - Perturbations from other gravitational bodies
    - Gravitational forces due to non-sphericity of the central body
    - General relativistic effects
  - Nongravitational forces on the spacecraft
    - Thruster firings
      - Trajectory control
      - Attitude control
    - Gas leaks
    - Solar radiation pressure
    - Aerodynamic drag

Example Trajectory: Earth-Mars

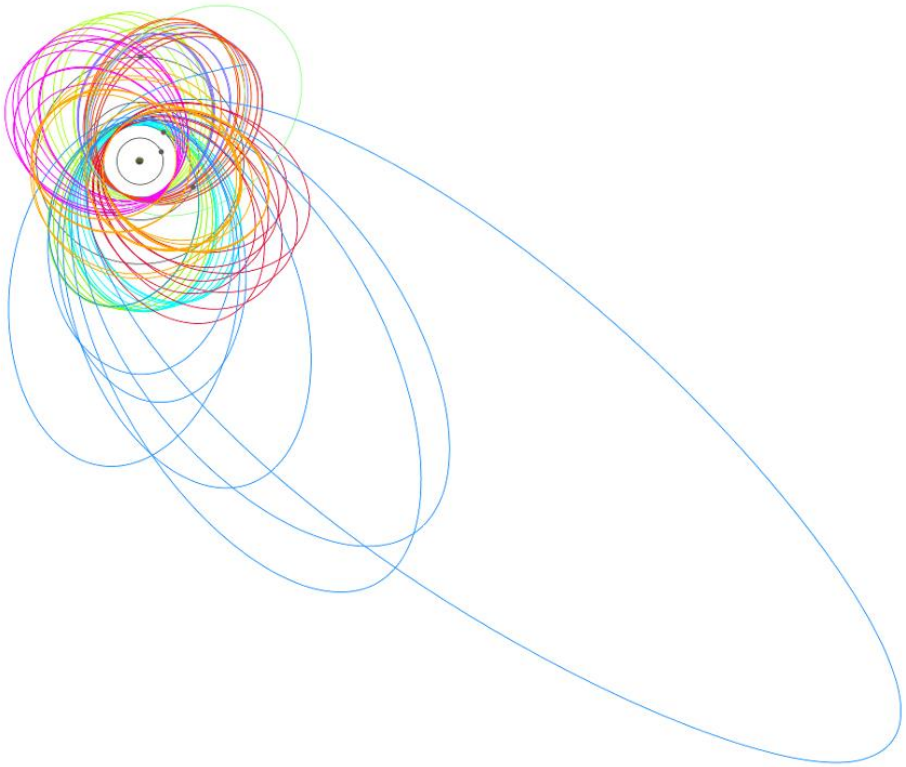


- **For many interplanetary missions, the trajectory plan has three phases:**
  1. Escape the Earth's gravitational influence
  2. Complete an interplanetary transfer orbit
  3. Flyby a target body, enter a planned orbit at the target body, or land on the target body

Example Trajectory: Europa Clipper Mission Jovian Tour



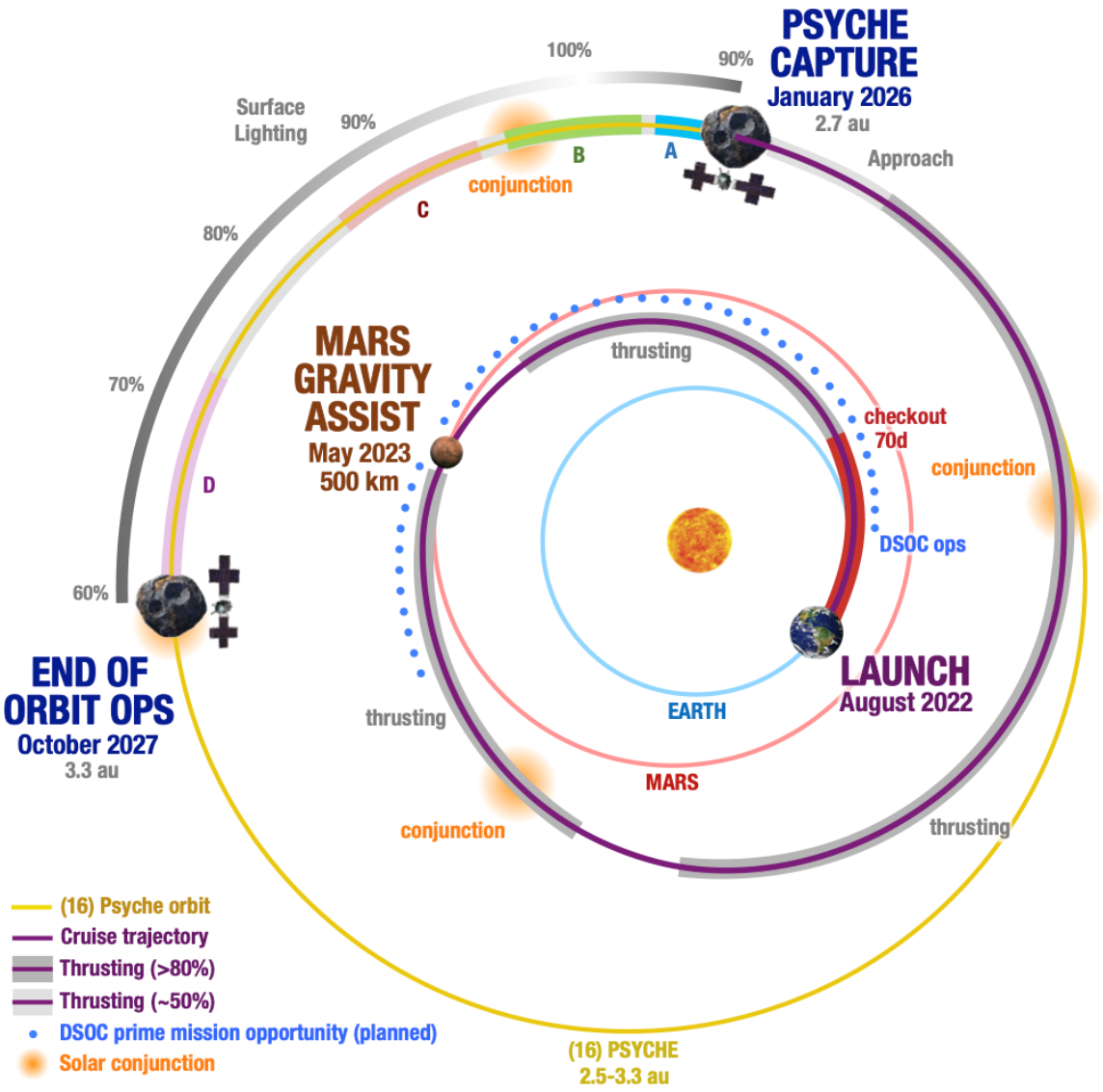
Interplanetary transfer



- 70 satellite flybys (51 Europa, 12 Ganymede, 7 Callisto
- Lowest flyby – 25 km altitude at Europa
- Flyby ground speed of 4.4 km/s

Source: Europa Clipper Navigation Team

Example Trajectory: Psyche Low Thrust Transfer to Asteroid



Source: Psyche Navigation Team

Navigation

- **Flying the spacecraft from launch to end of mission**
  - Reconstruction of position and velocity up to current time (orbit determination)
  - Predict future path of spacecraft
  - Compare actual course with planned course, and make course adjustments as necessary (flight path control)
- **Orbit determination**
  - Use tracking data to compute spacecraft's current trajectory
  - Radiometric data types
    - Doppler - measures line-of-sight velocity of spacecraft relative to tracking station
    - Range - measures line-of-sight distance of spacecraft relative to tracking station
    - Delta Differential One-way Range (DDOR) - measures plane-of-sky angle between spacecraft and a baseline between two tracking stations
  - Optical data
    - Uses onboard camera to measure angle between spacecraft and target body
- **Flight Path Control**
  - At predetermined times in the mission, compare predicted course with actual course
  - If outside of tolerance, compute maneuver to re-target
  - Can optimize current and future maneuvers to minimize fuel usage
  - Keep track of fuel usage

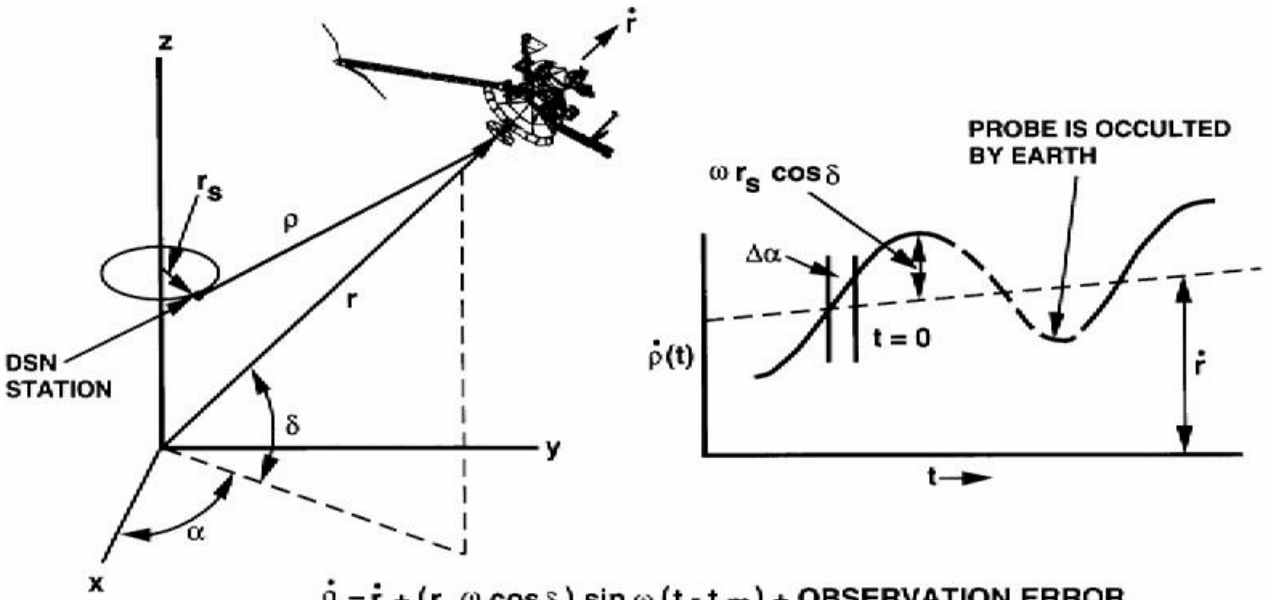
Navigation Data

- **Navigation data are scalar variables (e.g., range, range-rate, angle)**
- **The state of the spacecraft, however, is a vector (3 components of position and velocity each in a Cartesian coordinate system)**
  - Typically, additional variables also need to be estimated as part of the orbit determination process
    - Examples are components of thrust, solar radiation pressure parameters, parameters which describe a planet's gravity field
- **Doppler and range are the primary data types used, augmented by Delta-DOR and/or optical data types at certain phases of the mission**
- **Amount of tracking data varies greatly depending on mission phases and requirements**
  - Critical times (launch, approach or flyby, maneuvers) requires nearly continuous levels of tracking
  - Quiet cruise phases can use as little as 1 pass of data every few weeks

Doppler and Range Data

- Doppler measurements are made when a single tracking station radiates a signal to a S/C which in turn multiplies the received signal by a constant (turn-around ratio) and sends the signal back to the transmitting station. The signal frequency is Doppler shifted on both the up and down-link paths.
- Range measurements are the round-trip light time for a signal to propagate between a ground station and S/C and measures the line-of-sight component of the S/C position. The ranging signal consists of a sequence of sinusoidal tones phase modulated on the carrier.

- **TWO-WAY RANGE AND DOPPLER DIRECTLY MEASURE LINE-OF-SIGHT COMPONENTS OF SPACECRAFT STATE**
- **DIURNAL SIGNATURE OF EARTH ROTATION ALSO PROVIDES ANGULAR STATE INFORMATION**



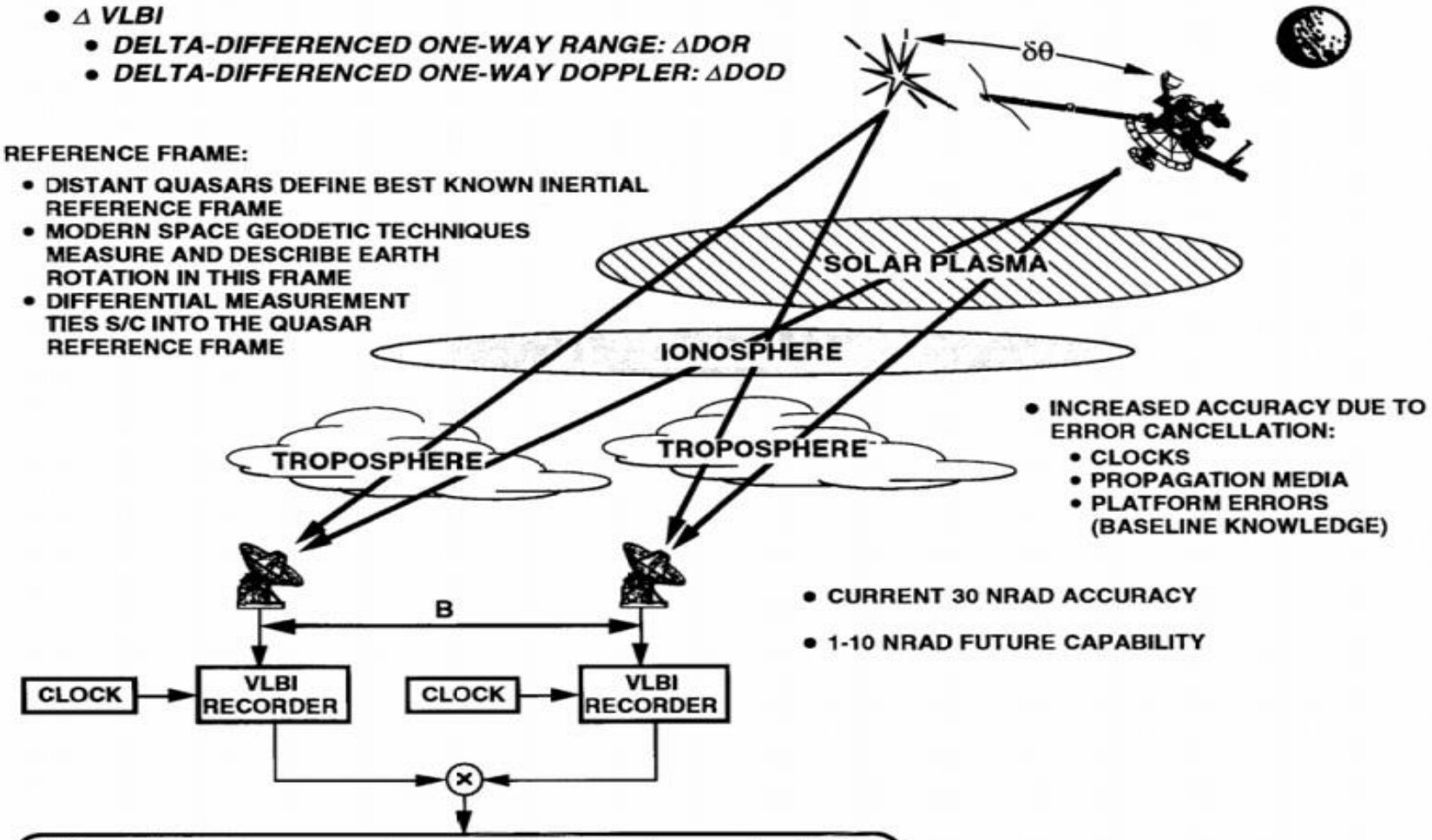
$$\dot{\rho} = \dot{r} + (r_s \omega \cos\delta) \sin\omega (t - t_m) + \text{OBSERVATION ERROR}$$

WHERE t_m IS THE TIME OF MERIDIAN CROSSING

Delta-DOR

- *Δ VLBI*
  - *DELTA-DIFFERENCED ONE-WAY RANGE: ΔDOR*
  - *DELTA-DIFFERENCED ONE-WAY DOPPLER: ΔDOD*

- REFERENCE FRAME:
  - DISTANT QUASARS DEFINE BEST KNOWN INERTIAL REFERENCE FRAME
  - MODERN SPACE GEODETIC TECHNIQUES MEASURE AND DESCRIBE EARTH ROTATION IN THIS FRAME
  - DIFFERENTIAL MEASUREMENT TIES S/C INTO THE QUASAR REFERENCE FRAME



- INCREASED ACCURACY DUE TO ERROR CANCELLATION:
  - CLOCKS
  - PROPAGATION MEDIA
  - PLATFORM ERRORS (BASELINE KNOWLEDGE)

- CURRENT 30 NRAD ACCURACY
- 1-10 NRAD FUTURE CAPABILITY

$$\Delta\rho_{s/c} = B \cos\theta_{s/c} + c\,(\Delta\tau_{clock} + \Delta\tau_{inst} + \Delta\tau_{media}) + NOISE$$

$$\Delta\rho_{Q} = B \cos\theta_{Q} + c\,(\Delta\tau_{clock} + \Delta\tau_{inst} + \Delta\tau_{media}) + NOISE$$

Clocks

- **Any error in the timetag of the observations will lead to an error in the range or range-rate measurements**
- **The Deep Space Network tracking stations are equipped with very stable atomic time and frequency reference systems**
  - Each deep space complex has four atomic frequency standard.
  - One of them at each site, usually a hydrogen maser, serves as the source for all coherent, precision frequencies and provides the reference for the station master clock.
  - Each site synchronizes its clock with the NIST realization of UTC via common-view GPS signals.

- **Spacecraft clocks are not as accurate, so we rely on 2-way data (the timetag is done at the station)**
- **New technologies, such as the Deep Space Atomic Clock (DSAC) can enable accurate 1-way tracking data**

Table 1. DSN Frequency Standard Performance.

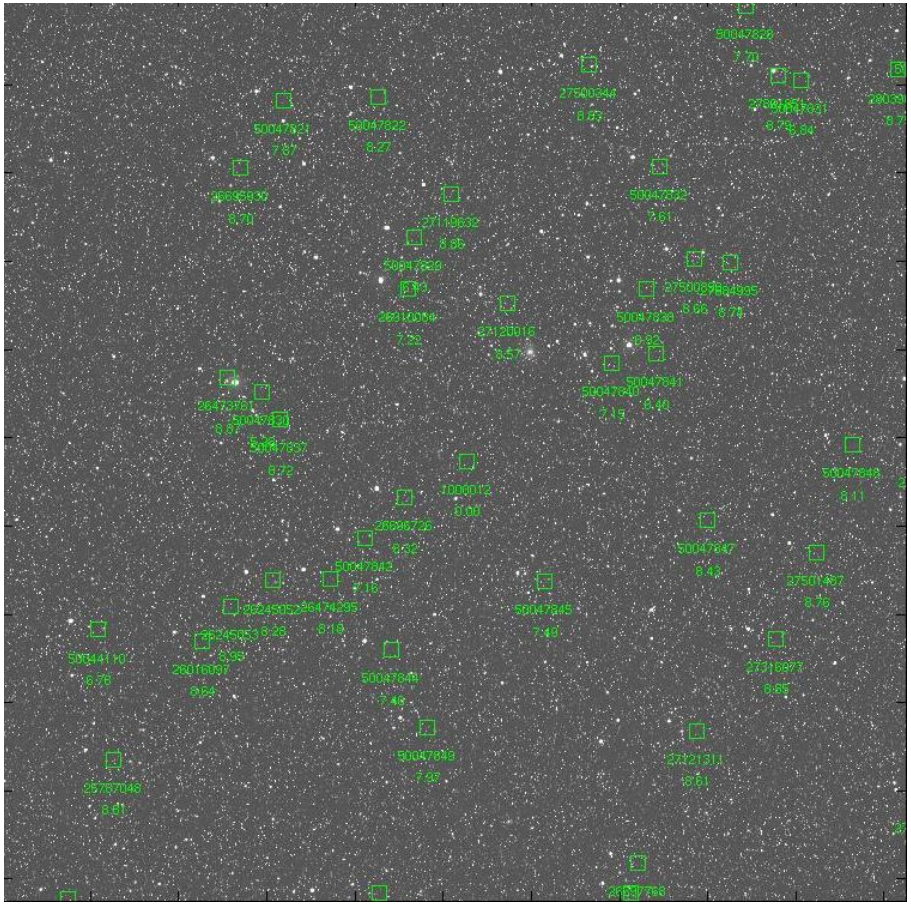
Parameter	Value
Frequency Offset Relative to UTC	
Worst Case	$<9 \times 10^{-13}$
Typical	$<3 \times 10^{-13}$
Resolution reconstructed by analysis (3 σ)	$<1 \times 10^{-13}$
Fractional Frequency Drift	
Specified	1×10^{-13} /10 days
Typical	$<3 \times 10^{-14}$ /10 days
Setability	10 ns
Offsets from UTC	
Requirement	< 3 μs
Resolution from UTC, reconstructed by analysis (3 σ)	< 1 μs

From DSN document 810-005 304A

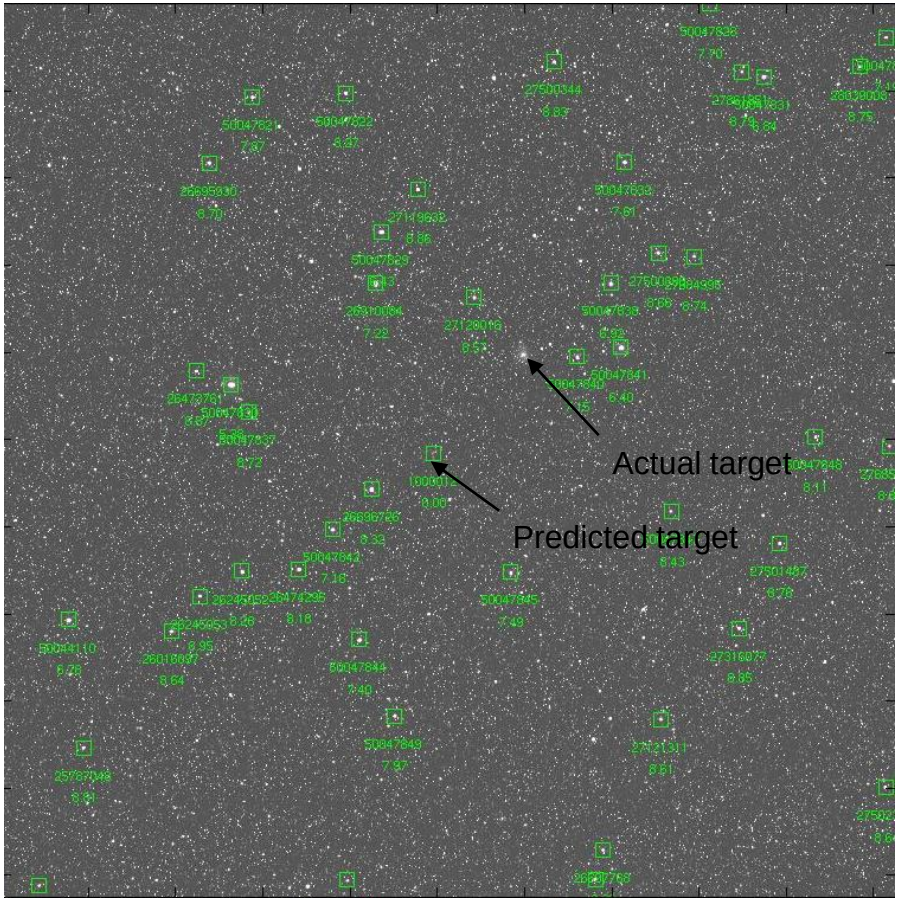
Optical Navigation

- **Use onboard camera to take images of natural bodies against star background**
  - Provides measurement of the angle between the spacecraft and target body in inertial reference frame
  - Stars, due to their large distances, provide accurate information about the orientation of the spacecraft
- **Accuracy of data type dependent on ability to locate center-of-figure of target body**
  - "Unresolved" (point source) target body locations can be determined to sub-pixel accuracy, yielding data accuracies on order of microradians
  - Extended bodies harder to determine centers and depends on whether good shape model exists
- **Opnav primarily used on approach to target and provides extremely good target relative information**
  - Especially useful for bodies whose ephemeris is not well known, e.g., asteroids, comets, planetary satellites
  - Also used extensively for orbiting small bodies as terrain relative navigation critical for proximity operations

Optical Navigation Example

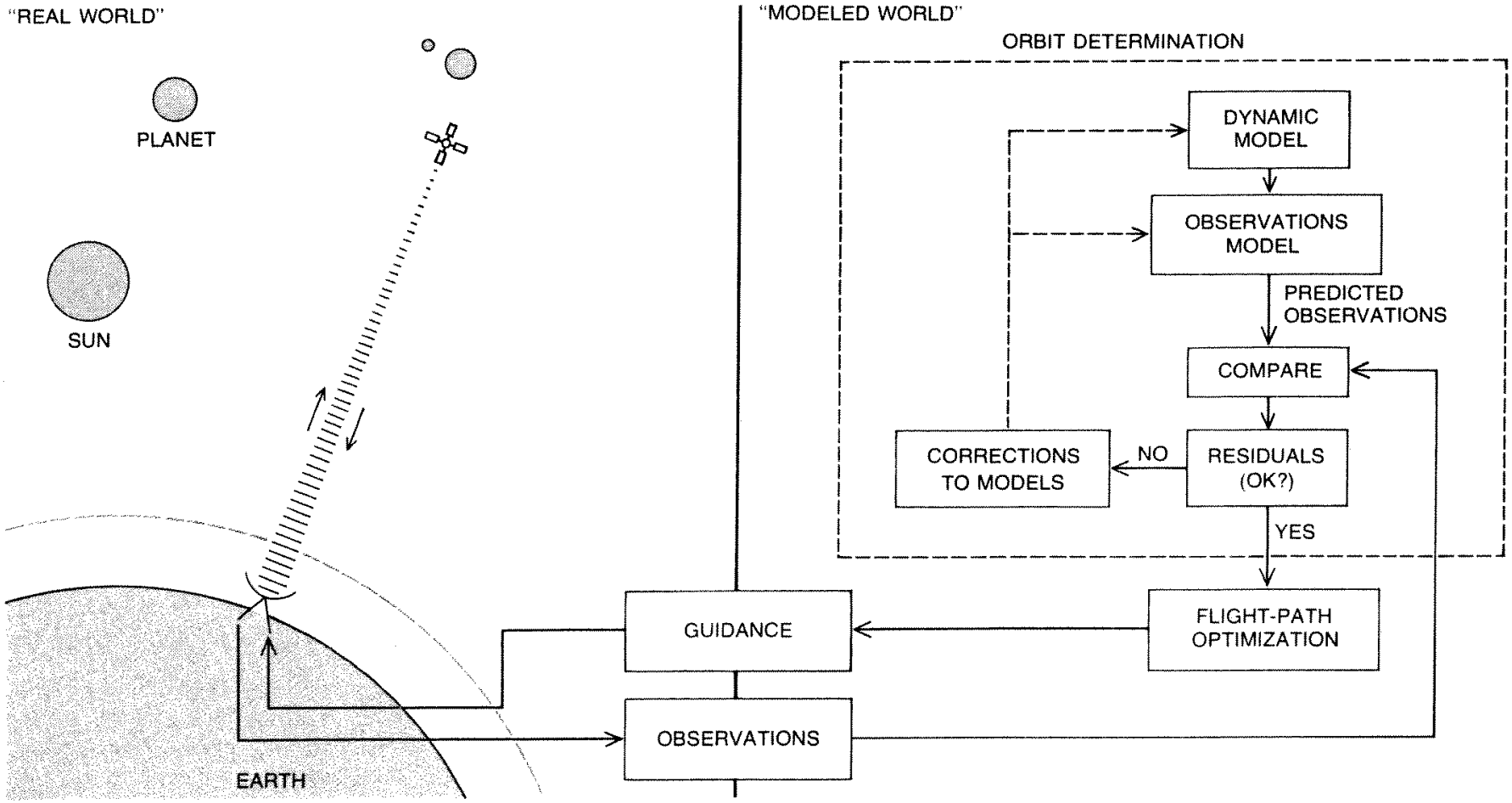


Before: stars and target not aligned with predict



After: stars aligned with predict, but not target

The Navigation Process



OD Solution Evaluation

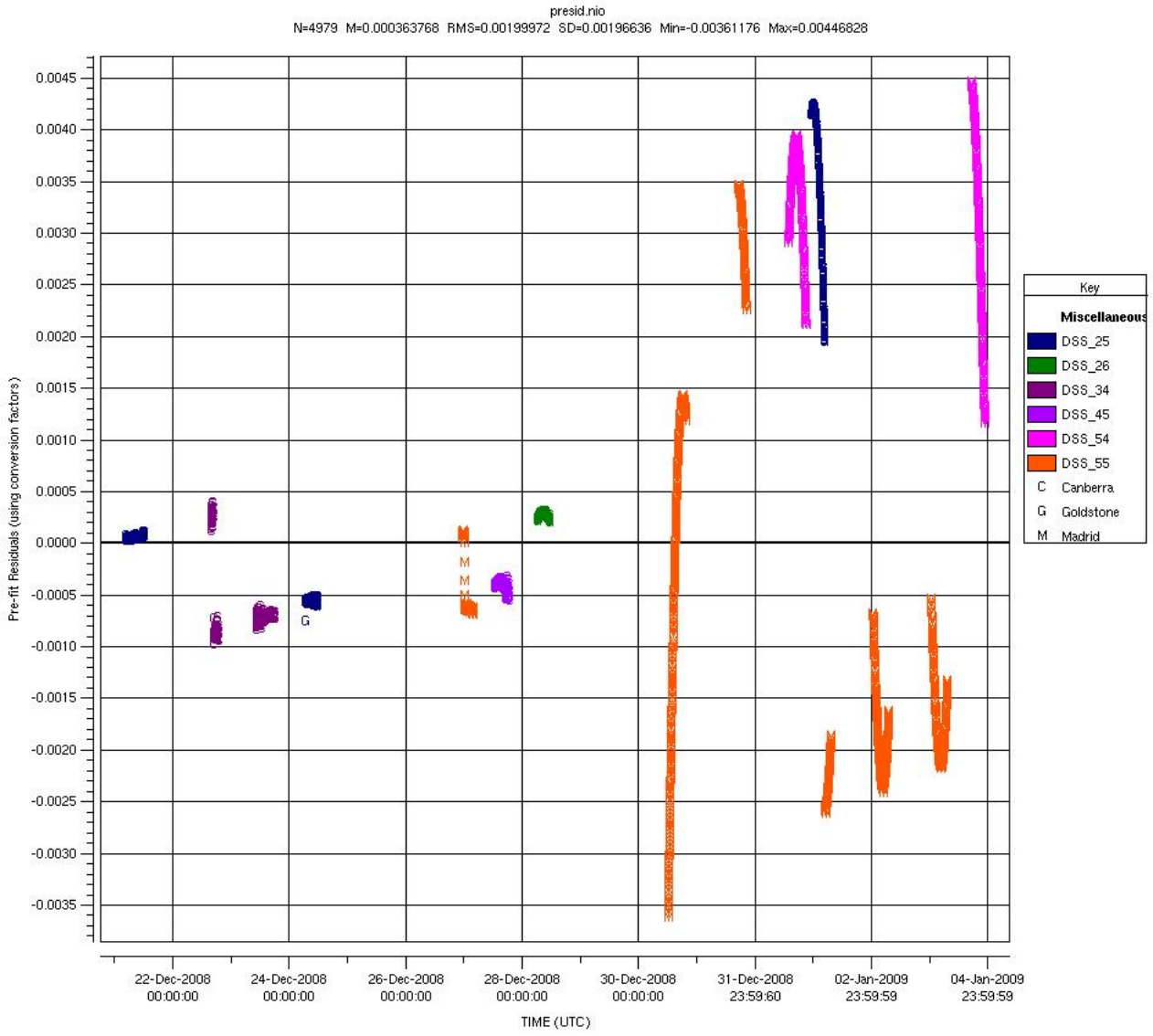
- **The residuals of the tracking data used in the solution should display no significant signatures or trends. The post-fit tracking data residuals should display a random scatter about a near zero mean.**
  - All residual outliers must be removed from the data set used in the solution.
  - The tracking data weights should not be less than the observed data scatter.

- **Are the estimated parameters a sufficient set to accurately model the trajectory and media? Are the data used in the solution sensitive to the dynamic and media model errors?**

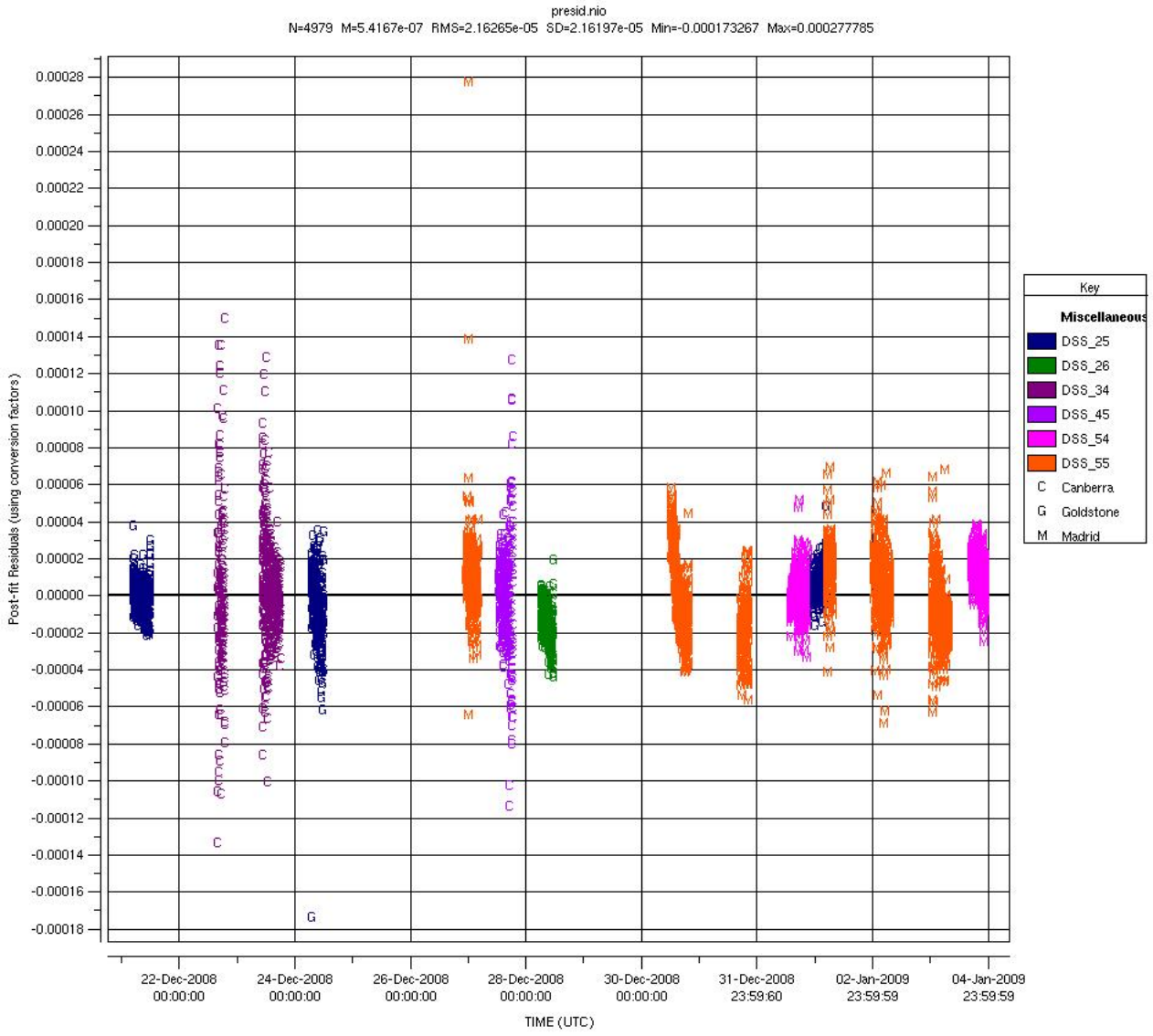
- **Do the corrections to the estimated parameters make physical sense (e.g. reflectivity coefficients should not solve out to a negative value)?**

- **The a priori uncertainties should be of appropriate magnitude. Parameter estimates are often sensitive to the a priori knowledge. Overly optimistic *a priori* knowledge can constrain the parameter estimate leading to a poor trajectory update.**

Orbit Determination – Pre-fit Doppler Residuals

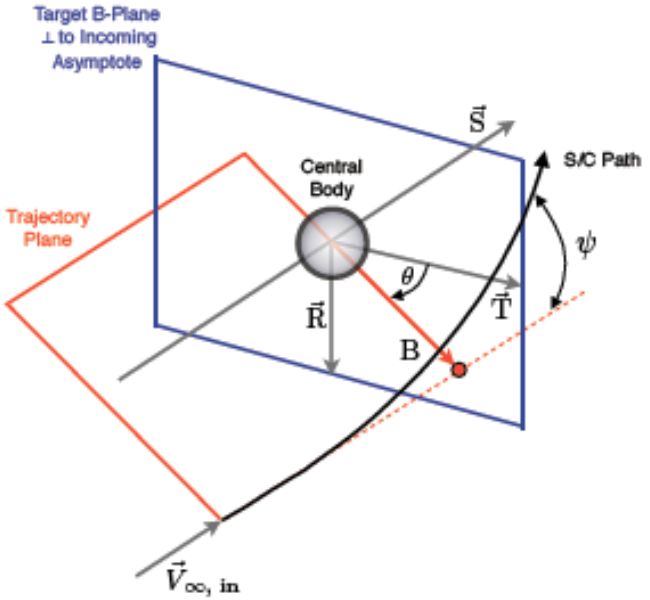


Orbit Determination – Post-fit Doppler Residuals

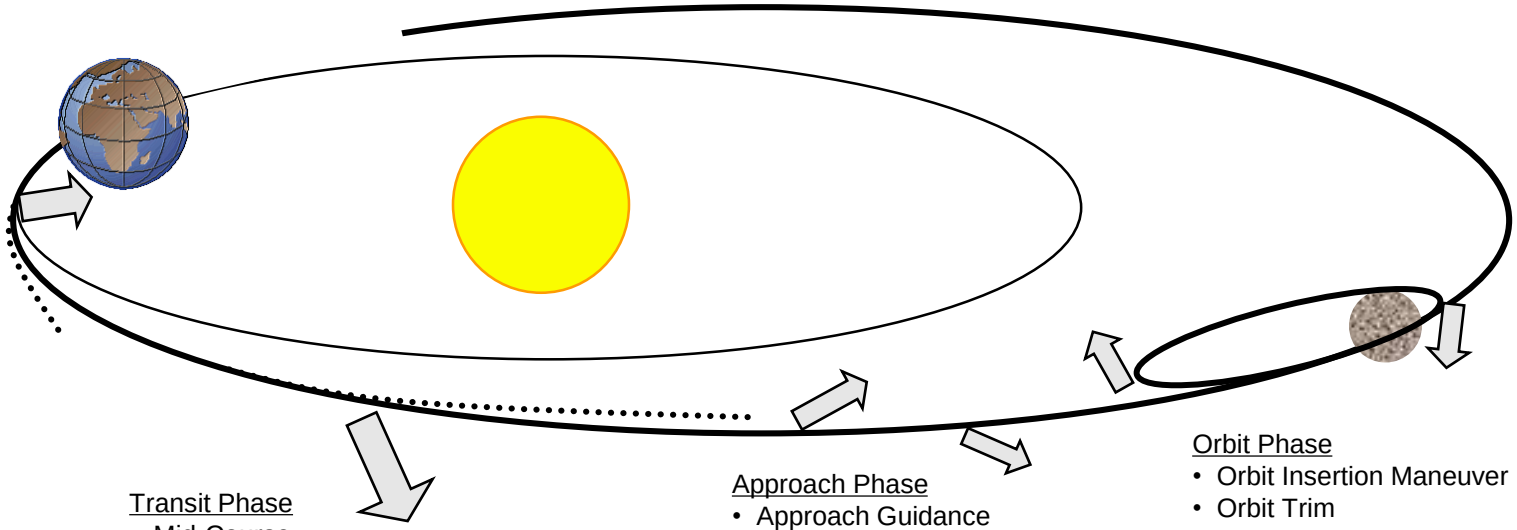


Solution Mapping

- **Spacecraft solution estimates typically done in Cartesian coordinate system, centered on primary gravitationally attracting body**
- **For assessing results, solution always mapped from epoch state to time and coordinate frame of interest**
- **Standard for interplanetary trajectories is the B-plane at planetary encounter**
  - Used for flybys, orbit insertions
- **Other examples might be equator longitude crossings or orbit periods for planetary orbiters**
- **Waypoints for small body proximity operations**

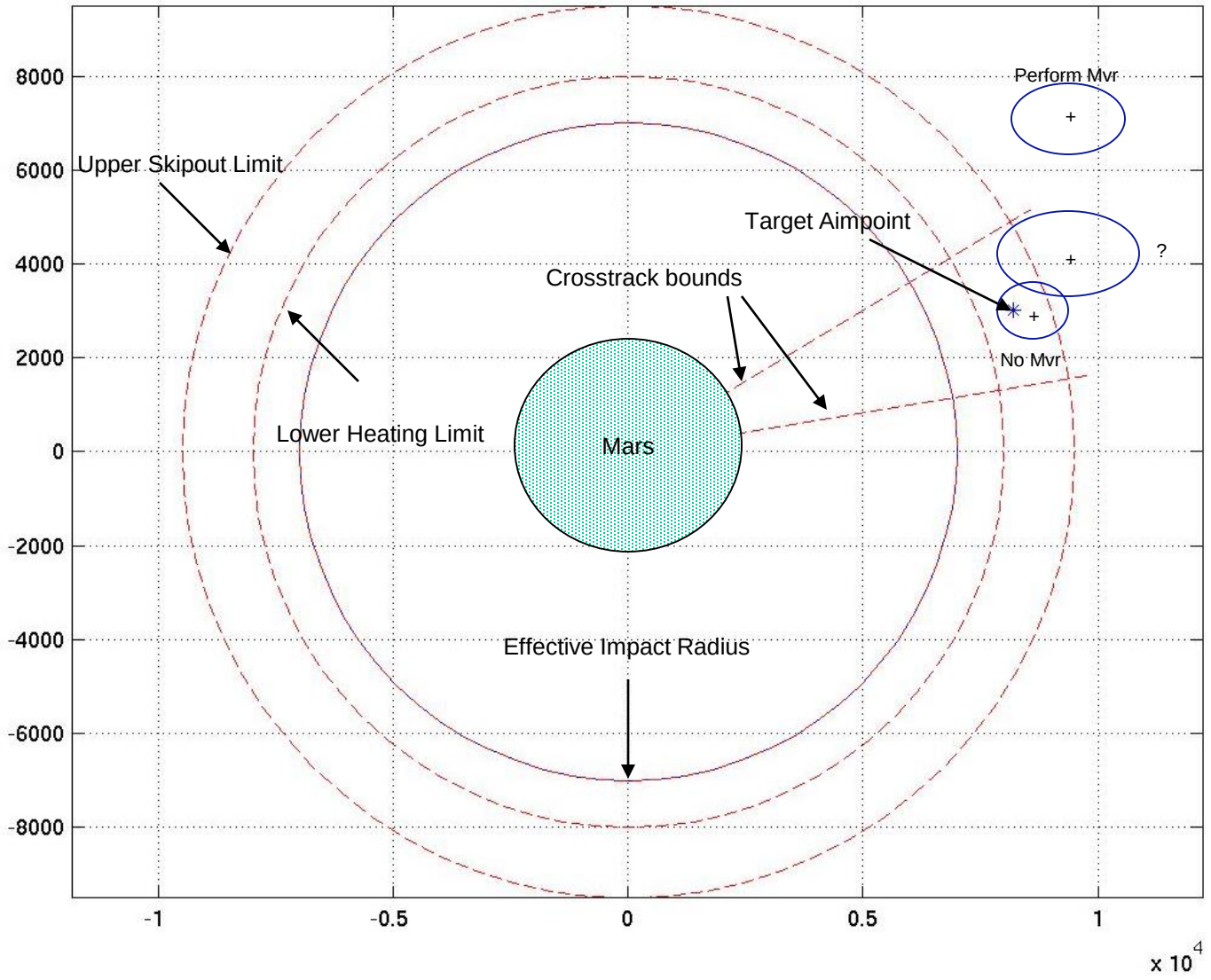


Closing the Loop - Maneuver Planning and Design



- **Targets come from the baseline trajectory**
  - Baseline trajectory determined from a trajectory optimization
- **Orbit determination provides best estimate current position, velocity, and model parameters**
- **Maneuver team designs TCM(s) to guide the spacecraft to the targets. Design depends on:**
  - Whether more than one maneuver is being designed simultaneously (maneuver optimization)
  - Mode of maneuver (vector or turn-burn-turn)
    - Maneuver design may rely on spacecraft attitude
- **Maneuver delivery statistics are computed from maneuver execution errors and orbit determination error statistics**

Maneuver Planning and Design Example

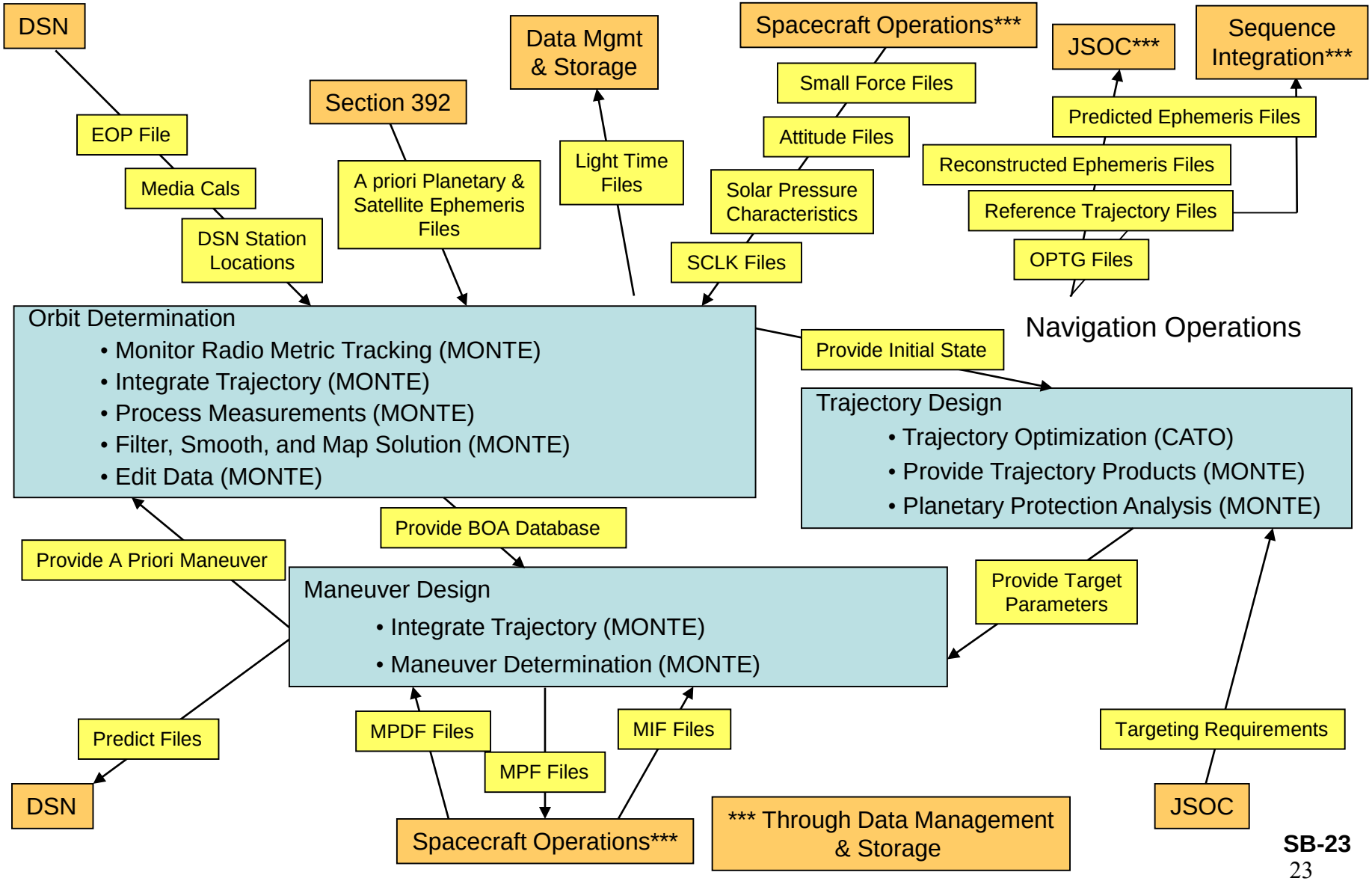


Blue ellipses represent orbit solutions propagated to to encounter

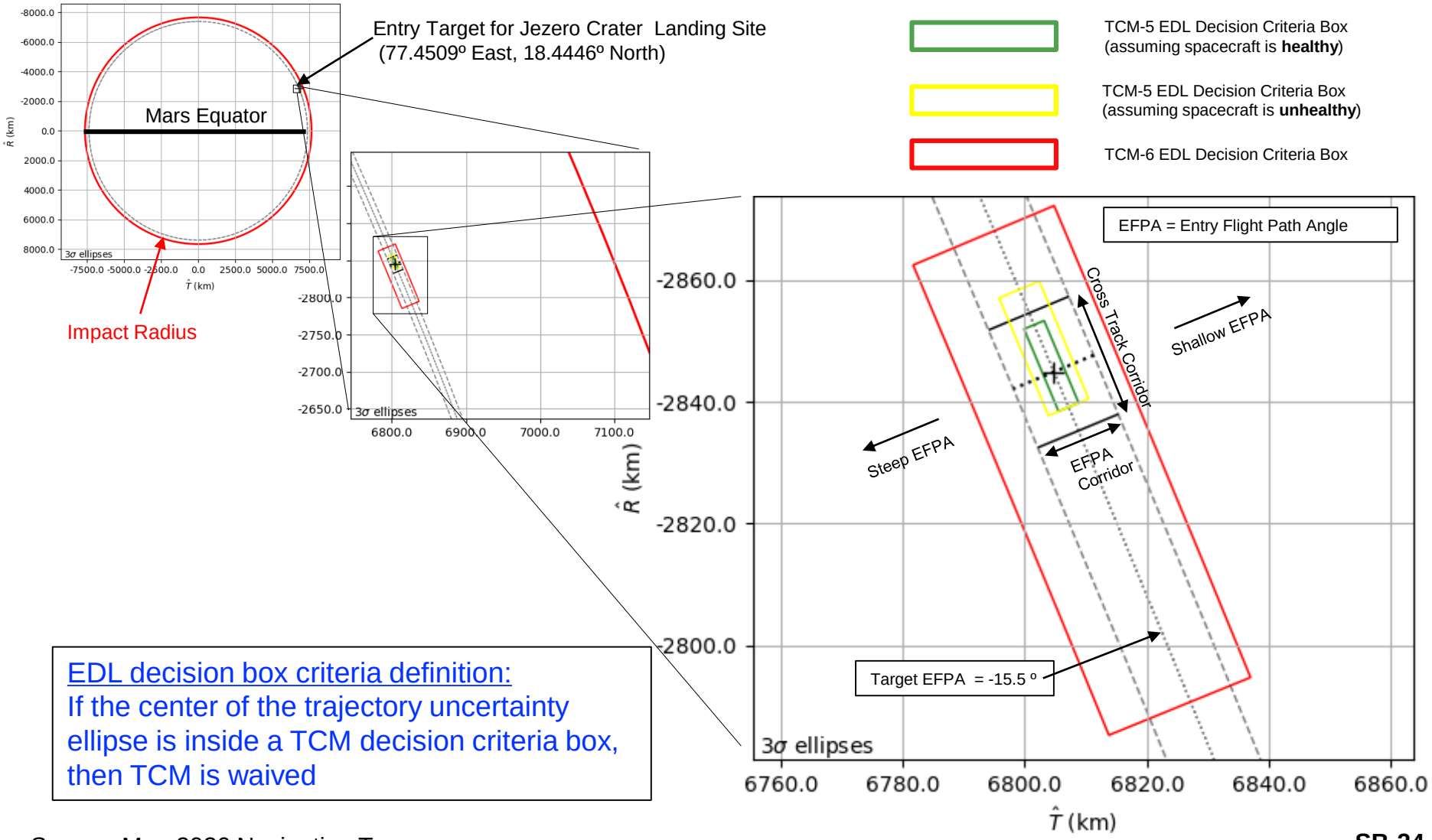
Ephemeredes and Gravity Models

- **In order to navigate to Solar System bodies, also need to know where the bodies are in heliocentric space**
- **Planetary/Satellite/Small Body ephemeredes are developed at JPL**
  - Word-wide Service
  - Continuous long term activity
  - Orbits are refined using measurements from variety of sources
    - Radar measurements from Earth
    - Astrometric images
    - Radio signals from spacecraft near planet of interest
    - Specialized techniques used for each ephemeris class
- **Gravity Models**
  - World wide standard
  - Updated and refined based on best data from orbiters and/or flybys

Navigation Software Flow



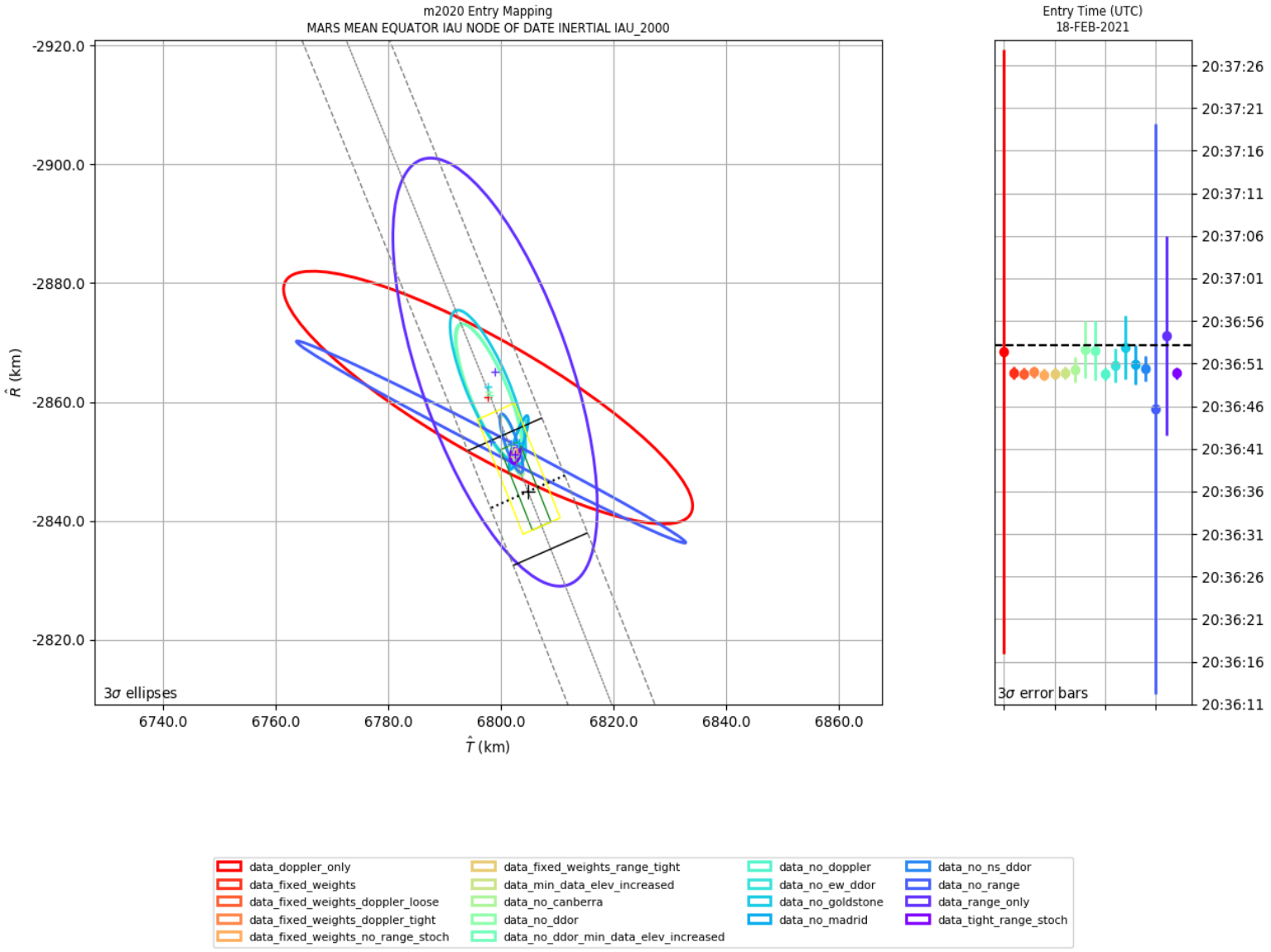
Example: Mars 2020 Target In The B-plane For Jezero Crater Landing Site



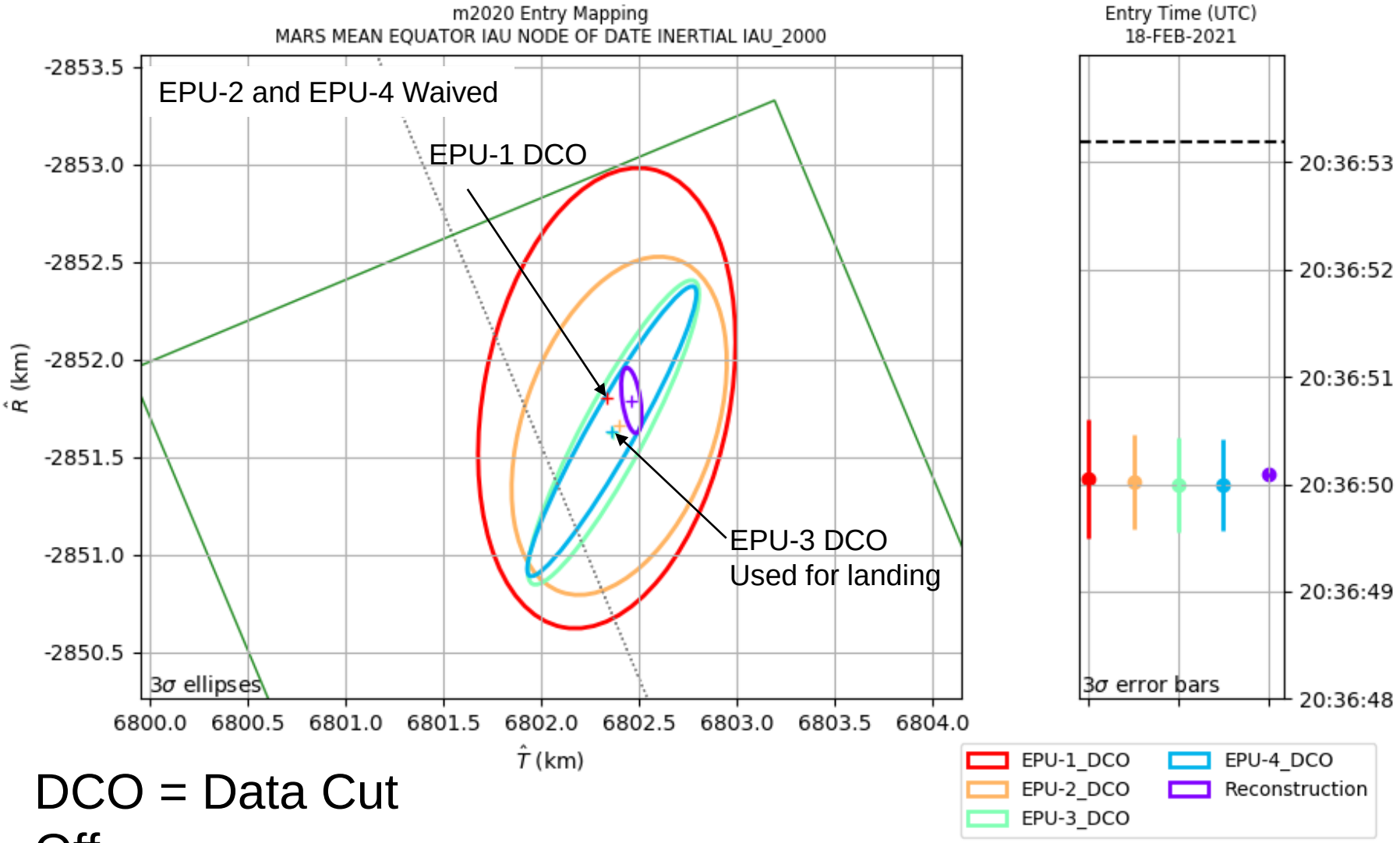
EDL decision box criteria definition:
If the center of the trajectory uncertainty ellipse is inside a TCM decision criteria box, then TCM is waived

Source: Mars2020 Navigation Team

Example: Mars2020 B-plane at E-10 days



Example: Mars2020 Final EPU solutions



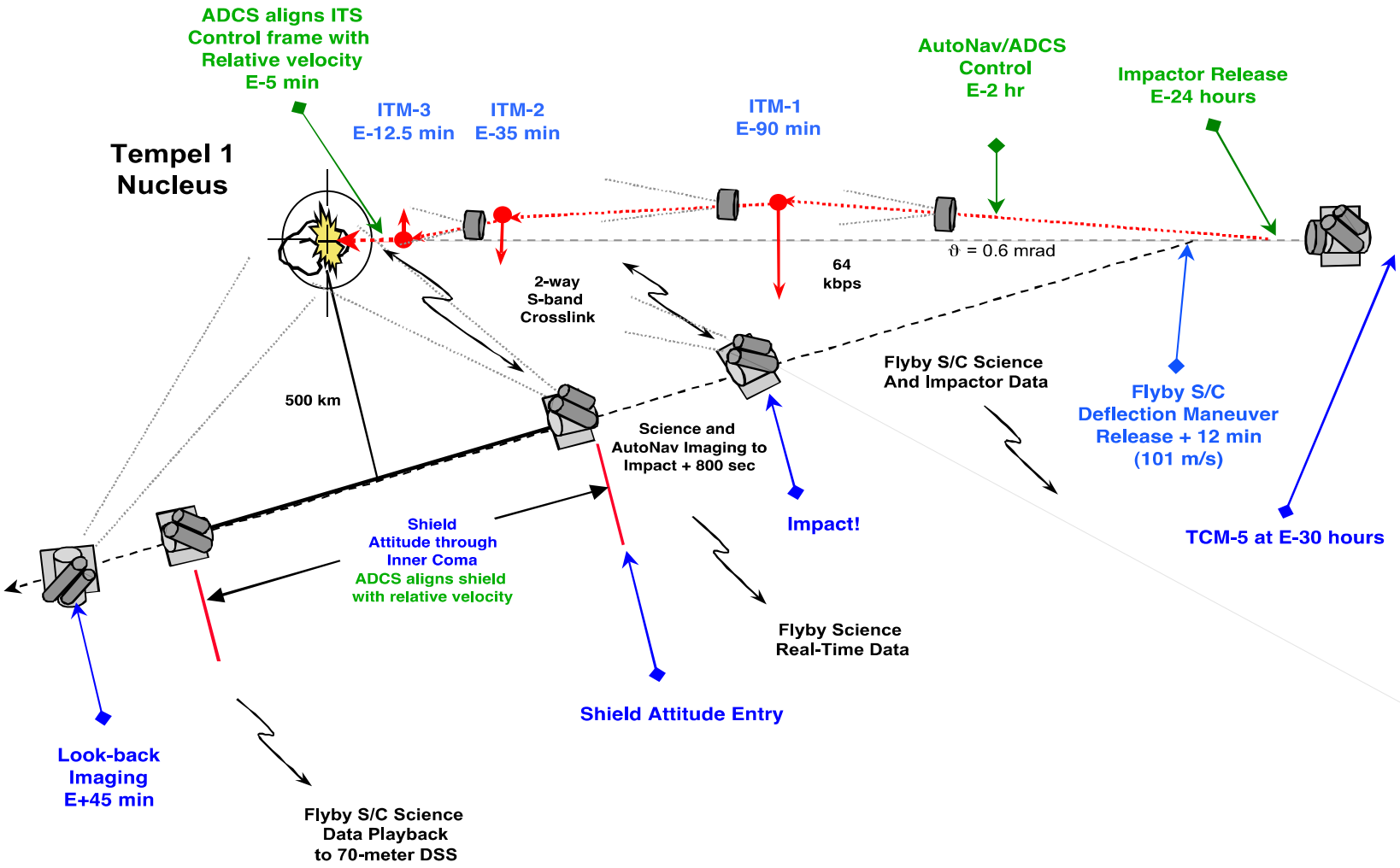
DCO = Data Cut Off

Source: Mars2020 Navigation Team

Autonomous Navigation

- **For some mission scenarios, ground in the loop navigation may not be capable of achieving mission objectives due to round-trip light time delays and tight navigation requirements**
- **Place certain computational elements of navigation onboard a spacecraft so it can compute its own orbit and maneuvers to achieve desired target conditions**
- **Current versions of Autonav is based solely on optical data**
  - Entirely self-contained, without need for uplink or downlink to compute solutions
    - Unlike radio data, does not require the use of DSN antennas
    - Does not depend on Earth-based parameters which need to be updated, such as media calibrations, Earth orientation parameters
- **Has been used on 4 missions to date, and one planned in 2022**
  - Deep Space 1- 1st demonstration of onboard Autonav, used for portions of cruise and flyby of comet Borrelly (JPL)
  - Stardust – used to track comet Wild 2 during flyby (JPL)
  - Deep Impact – used for targeting impactor and tracking impact on flyby spacecraft (JPL)
  - OSIRIS-REx – to achieve meter level landing accuracy on asteroid Bennu (GSFC/LM)
  - DART – terminal guidance to impact secondary asteroid of Didymos binary system (APL)

Example: Deep Impact Mission



Deep Impact - Impactor



Conclusion

- **Deep space navigation techniques presented here have been used for almost 60 years**
- **Used to navigate missions to all planets in the Solar System, plus asteroids, comets, and Kuiper Belt Objects**
- **Techniques were pioneered at JPL, and with only minor modifications, are used by space agencies around the world to navigate deep space missions**
- **Future**
  - Future changes are likely to be evolutionary, rather than revolutionary
  - More use of autonomy, 1-way radiometric using DSAC or equivalent
  - Some different techniques are theoretically possible (pulsar-based navigation, stellar navigation), but probably will not be used practically for some time to come